

# Carl Ziegler

**Associate Professor (tenured) • Director, SFA Observatory**

Department of Physics, Engineering, and Astronomy

Stephen F. Austin State University • Nacogdoches, Texas

[carlziegler@gmail.com](mailto:carlziegler@gmail.com) • [carlziegler.space](http://carlziegler.space)

---

Observational astrophysicist studying binary and multiple stars and their influence on planet detection, formation, and architecture, combining TESS, Gaia, and high-resolution imaging. Founder and lead of the SOAR TESS Survey (2,982 targets) and lead analyst of the Robo-AO Kepler Survey (3,857 systems); Co-Investigator on a NASA TESS Cycle 9 program selected for funding; background in adaptive-optics instrumentation. At SFA since 2020: observatory director, research mentor to 28 undergraduates, public-observing organizer, and instructor from introductory astronomy through upper-level astrophysics and physics (4.59/5 mean rating across 33 sections).

[RESEARCH, p. 1](#) • [MENTORING & TEACHING, p. 3](#) • [OBSERVATORY & OUTREACH, p. 4](#) • [PUBLICATIONS, p. 6](#)

## Academic Appointments

---

Aug. 2026 **Associate Professor, with tenure**

–present Stephen F. Austin State University

Sept. 2020 **Assistant Professor**

–July 2026 Stephen F. Austin State University

2020–present **Director, SFA Observatory**

2018–2020 **Dunlap Postdoctoral Fellow**

Dunlap Institute for Astronomy and Astrophysics, University of Toronto.

## Education

---

2018 **Ph.D., Physics**, University of North Carolina at Chapel Hill.

Dissertation: *Characterization of Exoplanets and Stellar Systems with New Robots*. Advisor: Nicholas Law.

2013 **M.S., Physics**, Southern Illinois University Carbondale.

Thesis: *Adsorption of Neon on Open Carbon Nanohorn Aggregates*. Advisor: Aldo Migone.

2009 **B.A., Physics and Mathematics**, William Jewell College.

Undergraduate research: variable stars and globular clusters. Advisor: Maggie Sherer.

## Research Program

---

2018–present **Founder and Lead, SOAR TESS Survey.** Conduct HRCam speckle imaging on the 4.1-m SOAR telescope with Andrei Tokovinin and colleagues. Set target selection and observing strategy; develop Python analysis workflows; lead survey papers and undergraduate research. Survey III incorporates observations through January 2024, combining detection limits and Gaia information to measure close-companion demographics.

2013–2018 **Lead Analyst, Robo-AO Kepler Survey.** With Nicholas Law and Christoph Baranec, led four first-author studies of companion detection, planetary-radius corrections, physical association, and Gaia close-binary recoverability. Survey imaging supported Kepler and K2 planet validation.

2019–present **One-Hit Wonders and TESS Single-Transit Follow-up.** Developed target-selection, period-prediction, and photometry workflows for the 0.5-m robotic telescope at Deep Sky West, New Mexico; observing began in September 2019. SFA students use campus photometry and public TESS data to search for subsequent transits.

## Current Directions and Community Data

Current work combines speckle imaging, Gaia astrometry, and population inference to study the close-binary deficit among TESS hosts, with undergraduate participation. The selected NASA program extends companion characterization to dual-band Robo-AO-2 imaging; small-telescope photometry supports long-period planet follow-up.

SOAR imaging contributes to the TESS Follow-up Observing Program (SG3) and ExoFOP-TESS; published SOAR I and II companion catalogs are available through VizieR.

## Research Support and Observing Time

- 2026 selection **Co-Investigator and SFA Subaward Lead.** NASA TESS Cycle 9 General Investigator Program, *Resolving the Multiplicity Bottleneck: Dual-Band Imaging of TESS Planetary Environments* (program PI: Christoph Baranec, University of Hawai'i). Selected for funding in 2026; Phase 2 budget submitted. Planned period: February 2027–January 2029. Responsibilities: TESS target coordination, imaging analysis and interpretation, and undergraduate research mentoring.
- 2025–2026 **Faculty Mentor, SFA Summer Undergraduate Research Experience (SURE).** Competitive internal funding for student and faculty research: Gerson Bruton<sup>†</sup> (2025); Seth Batson<sup>†</sup> (2026).
- 2020B–2022A **Principal Investigator, SOAR Telescope Allocations.** Selected awards for TESS planet-host characterization: 2022A (3 nights), 2021A (4 nights), and 2020B (3 nights).

## Selected Publications

<sup>†</sup> SFA undergraduate research mentee; <sup>‡</sup> Toronto summer undergraduate mentee. Full bibliography and authorship summary begin on p. 6.

- 2026 **Ziegler, C.**, Tokovinin, A., Briceño, C., Law, N. M., Mann, A. W., & Batson, S.<sup>†</sup> *SOAR TESS Survey III: Mapping the Stellar Companion Deficit around TESS Planet Candidates*. [arXiv:2608.26250](https://arxiv.org/abs/2608.26250); submitted to *AJ*. Lead-author analysis of the close-companion deficit among 1,199 dwarf primaries, accounting for imaging sensitivity.
- 2022 Salama, M., **Ziegler, C.**, et al. *An Adaptive Optics Census of Companions to Northern Stars Within 25 pc with Robo-AO*. *AJ*, **163**, 200. Second-author contribution to a volume-limited census of nearby stellar multiplicity.
- 2021 **Ziegler, C.**, Tokovinin, A., Latiolais, M.<sup>‡</sup>, Briceño, C., Law, N., & Mann, A. W. *SOAR TESS Survey. II. The Impact of Stellar Companions on Planetary Populations*. *AJ*, **162**, 192. Lead-author survey paper linking close stellar companions to planetary populations, with an undergraduate coauthor.
- 2020 **Ziegler, C.**, Tokovinin, A., Briceño, C., Mang, J.<sup>‡</sup>, Law, N., & Mann, A. W. *SOAR TESS Survey. I. Sculpting of TESS Planetary Systems by Stellar Companions*. *AJ*, **159**, 19. Established the survey's high-resolution census and companion constraints for TESS planet-candidate hosts.
- 2018 **Ziegler, C.**, et al. *Measuring the Recoverability of Close Binaries in Gaia DR2 with the Robo-AO Kepler Survey*. *AJ*, **156**, 259. Quantified Gaia's close-binary completeness using resolved Robo-AO imaging.
- 2018 **Ziegler, C.**, et al. *Robo-AO Kepler Survey. V. The Effect of Physically Associated Stellar Companions on Planetary Systems*. *AJ*, **156**, 83. Distinguished bound companions from chance alignments to test their relationship to planetary systems.
- 2018 **Ziegler, C.**, et al. *Robo-AO Kepler Survey. IV. The Effect of Nearby Stars on 3857 Planetary Candidate Systems*. *AJ*, **155**, 161. Survey-wide companion analysis and corrections to planet radii inferred from blended photometry.

SOAR imaging contributions include TOI-1338, TESS's first transiting circumbinary planet (Kostov et al. 2020); TIC 278956474, a young quadruple with two eclipsing binaries (Rowden et al. 2020); and TOI-700, the first habitable-zone Earth-size TESS planet (Gilbert et al. 2020).

## Undergraduate Research Mentoring

2020–2026 28 undergraduate research mentees at SFA since 2020. Projects span exoplanet photometry, stellar multiplicity, single-transit recovery, Python analysis pipelines, high-resolution imaging, and astro-physical simulations.

Earlier summer mentoring at Toronto included SURP students James Mang<sup>‡</sup> and Julia Pasiecznik<sup>‡</sup> (University of Toronto).

## Student Publications, Presentations, and Outcomes

- **Student coauthors.** James Mang<sup>‡</sup>: fourth author, SOAR TESS Survey I (*AJ*, 2020). Madelyn Latiolais<sup>†</sup>: third author, SOAR TESS Survey II (*AJ*, 2021). Seth Batson<sup>†</sup>: coauthor, SOAR TESS Survey III (submitted, 2026).
- **Student research presentations, selected examples.** Madelyn Latiolais<sup>†</sup>: Keck-imaging poster (SURE, 2020); Alyssa Auten<sup>†</sup>: *The Search for Exoplanets: TESS*, Joint Spring Meeting: TSAAPT/TSAPS/SPS Zone 13 (March 2023); Seth Batson<sup>†</sup>: *Atmospheric Analysis of a Hot Jupiter*, SURE poster (2026).
- **Funded summer research.** Gerson Bruton<sup>†</sup> developed the early Gravitas black-hole simulation (SURE, 2025). Seth Batson<sup>†</sup> analyzed JWST observations of WASP-39 b with Eureka! and POSEIDON, reproducing a published transmission-spectrum analysis (SURE, 2026).
- **Postgraduate outcomes.** REUs at the University of Hawai‘i and University of Chicago; graduate study at UT Austin and UT Dallas; two flight controllers supporting NASA mission operations; an engineer at Samsung Austin Semiconductor.
- **Professional preparation.** Guide scientific writing and presentations, graduate applications, interviewing, and research-project planning.

## Graduate and Interdisciplinary Mentoring

Supervised Katie Bailey’s M.S. Natural Sciences thesis (completed spring 2023), using automated TESS analysis to investigate exoplanet candidates. Thesis-committee service spans Natural Sciences, English, and Music, including Adam Miller’s Master of Composition project (completed fall 2025).

## Teaching and Curriculum Development

4.59/5 mean faculty-effectiveness rating across 33 sections, fall 2020–summer 2026 (mean of section averages; directed study and practicum excluded). Spring 2026: 4.62/5 in introductory astronomy and 4.82/5 in Observational Astronomy.

## Courses Taught at Stephen F. Austin State University

- **Observational astronomy:** Observational Astronomy (ASTR 3305); Advanced Observational Astronomy Laboratory (ASTR 3005).
- **Introductory astronomy:** Modern and Classical Astronomy (ASTR 1303), in person and online; Astronomy Laboratory (ASTR 1103).
- **Astrophysics:** Astrophysics (ASTR 3335); Astrophysics II, offered through independent study (PHYS 4175, spring 2024).
- **Physics:** Mathematical Methods in Physics (PHYS 3347).
- **Research and professional preparation:** Physics Practicum (PHYS 4170); Independent Study in Physics and Astronomy (PHYS 4175/4176); M.S. Thesis Research (PHYS 5389/5390).

## Course Design and Inclusive Practice

- **Observational astronomy.** Rebuilt the course around a semester-long research experience: students propose observations, collect and calibrate telescope data, evaluate uncertainty, and present findings. Authored the advanced-observing laboratory manual.
- **Introductory astronomy.** Designed online and face-to-face instruction with orientation modules, short concept videos, low-stakes checks, targeted mathematics practice, and formative feedback to support students with varied preparation. Integrated planetarium sessions and Stellarium observing activities.

- **Inclusive student support.** Through the Student Experience Project (2025), built belonging and growth-mindset practices into course design; set course policies that accommodate caregiving and work schedules and connect students with campus food, emergency-grant, and textbook resources.
- **Upper-level astrophysics.** Developed original notes, problems, examinations, and worked solutions; integrated independent investigations and discussion of current research papers.
- **Mathematical methods.** Taught the course three times, introducing Python and MATLAB exercises connecting Fourier analysis and related mathematics to physical problems.

## Teaching Resources and Professional Development

Completed ACUE pedagogy training (2024–2025) and SFA’s Certified Online Instructor program (2020). Share introductory course materials with departmental lecturers and coordinate observatory instruction for colleagues.

## Astronomy-Education Software

**Creator and Lead Developer, Gravitas** (2026). Open-source, browser-based astrophysics laboratory with 59 scenarios and 22 guided investigations linking orbital dynamics to observable signals. Grew from an undergraduate SURE simulation project in 2025 and was piloted in introductory astronomy; designed for access through Spanish localization, keyboard navigation, and offline use. [Live laboratory \(gravitas-sim.online\)](#) • [Code and instructor resources](#) • [doi:10.5281/zenodo.22800610](#) • [arXiv:2609.19327](#) (education preprint).

## Observatory Operations, Instrumentation, and Public Engagement

- 2020–present **Director, SFA Observatory.** Schedule observing for introductory and advanced astronomy labs, train student telescope operators, and coordinate facility access and weather-dependent programs. Research facilities include the 1.04-m (41-inch), 0.46-m (18-inch), and 0.41-m (16-inch) telescopes. Used all three for student research on exoplanet transits, asteroid searches and tracking, and eclipsing binaries; campus photometry also supports TESS single-transit follow-up. Arrange class visits and interdisciplinary use, including LeTourneau University astronomy classes and SFA astrophotography students.
- 2014–2018 **Adaptive-Optics Instrumentation.** During doctoral research at UNC–Chapel Hill, designed and built the Robo-SOAR/SRAO natural-guide-star adaptive-optics testbed, with reflective dual-knife-edge wavefront sensing and robotic control. Two lead-author SPIE papers (2016) document the optical/wavefront-sensor design and the Robo-AO survey. Optical and opto-mechanical design with Zemax and SolidWorks; experience with Shack–Hartmann, pyramid, and knife-edge wavefront sensing. Coauthor, Robo-AO-2 instrumentation proceedings (SPIE, 2026).
- 2021–present **Public Observing and Community Partnerships.** Since 2021, organized approximately 30 public observing nights (about three per semester), reaching an estimated 2,000 visitors. Coordinate student telescope operators and guides, ticketed attendance, and weather contingencies. Partner with regional schools, Scout troops, summer camps, university groups, and the Astronomical Society of East Texas; host observatory alumni nights supporting recruitment and student connections.
- 2023–2024 **Solar-Eclipse Programs.** Led the October 2023 annular-eclipse campus event for more than 200 attendees. Assisted with the April 2024 total-eclipse student trip and Nacogdoches library event; created an eclipse-science and safety video distributed through SFA channels.
- 2024 **SFA STEM Academy Curriculum.** Developed two 12-lecture curricula, on radiation and electromagnetism, with guided notes, activities, worksheets, solutions, and instructor resources for high-school teachers.

## Selected Invited and Contributed Presentations

- 2025 *Double Suns, Lonely Worlds: How Binary Stars Shape the Hunt for Exoplanets.* Invited colloquium, East Texas A&M University.
- 2024 *SOAR TESS Survey: Thousands of High-Resolution Images of Planet-Hosting Stars.* Fall joint meeting, Texas Sections of APS/AAPT and Zone 13 of SPS, Dallas, Texas.

- 2023 *SOAR TESS Survey: Thousands of High-Resolution Images of Planet Hosting Stars*. Joint Spring Meeting: TSAAPT/TSAPS/SPS Zone 13 (March 2023), Texas A&M University–Commerce.
- 2020 *SOAR TESS Survey: The Sculpting of Planetary Systems by Stellar Companions*. AAS 235, Honolulu, Hawai'i.
- 2019 *One-Hit Wonders: Hunting the Longest-Period TESS Planets*. TESS Science Conference I, Cambridge, Massachusetts; CASCA Annual Meeting, Montréal, Québec.
- 2019 *Death Stars? How Tight Binaries Impact TESS Planets with SOAR Speckle Imaging*. AAS 233, Seattle, Washington.
- 2018 *Robo-AO KOI Survey: LGS AO Imaging of Every Kepler Planetary-Candidate Host Star*. AAS 231, National Harbor, Maryland.
- 2017 *High-Resolution Imaging of Four Thousand Kepler Planetary-Candidate Host Stars*. Know Thy Star, Know Thy Planet, Pasadena, California.
- 2016 *The Robo-AO KOI Survey and the Development of a Southern Robotic AO System*. Invited seminar, Institute for Astronomy, Hilo, Hawai'i.

## Service

- 2023–present Member, College of Sciences and Mathematics College Council; college representative, University Grievance Board. Reviewed and approved more than 200 college curriculum changes.
- 2025 **Faculty Research Enhancement Grant Review Panel**. Evaluated 12 internal applications for scientific merit, budget feasibility, and external funding potential.
- 2024–present Member, Physics Degree Revision and Modification Committee. Helped develop a plan to join the Texas Physics Consortium and begin planning a Medical Physics degree track.
- 2022 **Chair, Expectations and Values Committee**. Led drafting of shared departmental professional expectations; presented the initial document for faculty feedback.
- 2021–present Member, Recruitment, Retention, and Outreach Committee.

## Professional Service

NASA proposal-review service: panelist on three grant-review committees during the SFA faculty appointment. Journal referee: *AJ*, *ApJ*, *MNRAS*, *PASP*, and *A&A*.

## Professional References

### Christoph Baranec

[baranec@hawaii.edu](mailto:baranec@hawaii.edu)

Astronomer, Institute for Astronomy, University of Hawai'i.

*Research collaborator; PI of the selected NASA TESS Cycle 9 program.*

### Nicholas Law

[nmlaw@physics.unc.edu](mailto:nmlaw@physics.unc.edu)

Associate Professor of Physics and Astronomy, University of North Carolina at Chapel Hill.

*Doctoral advisor and collaborator on the Robo-AO and SOAR surveys.*

### Suresh Sivanandam

[sivanandam@dunlap.utoronto.ca](mailto:sivanandam@dunlap.utoronto.ca)

Director and Associate Professor, Dunlap Institute, University of Toronto.

*One-Hit Wonders collaborator; colleague during the Dunlap Fellowship (2018–2020).*

## Complete Publications and Scholarly Outputs

Ordered by year, newest first. Author lists are abbreviated for long collaborations; **C. Ziegler** is shown in every entry. Journal versions replace their earlier preprints.

### Refereed Journal Articles

189 articles and reviews: 8 first-author, 6 second- or third-author, and 175 fourth-author or later. Published corrections are listed separately. The collaborative record includes exoplanet discovery and characterization studies using high-resolution imaging from the SOAR and Robo-AO surveys.

<sup>†</sup> SFA undergraduate research mentee; <sup>‡</sup> Toronto summer undergraduate mentee. Submitted work is listed separately from published journal articles.

#### 2026

- J189 Ferreira, T., Yana Galarza, J., Reggiani, H., et al. (including **Ziegler, C.**) (2026). Giant Planets and Eccentric Orbits are Common around Galactic Thick Disk Stars. *AJ*, **172**, 211.
- J188 Naponiello, L., Poretti, E., Rice, K., et al. (including **Ziegler, C.**) (2026). Refined parameters, formation, and dynamical stability of the wild exoplanet system K2-312 ≡ HD 80653. *A&A*, **713**, A140.
- J187 Rea, E., Günther, M. N., Dransfield, G., et al. (including **Ziegler, C.**) (2026). High five from ASTEP. Three validated planets and two eclipsing binaries in a diverse set of long-period candidates. *A&A*, **711**, A86.
- J186 Dransfield, G., Petit, A. C., Triaud, A. H. M. J., et al. (including **Ziegler, C.**) (2026). ASTEP confirmation of a pair of long-period Jupiter-sized planets with extremely low densities transiting TOI-791. *MNRAS*, **549**, *stag864*.
- J185 Eschen, Y. N. E., Wilson, T. G., Cameron, A. C., et al. (including **Ziegler, C.**) (2026). An ultra-short-period super-Earth and a sub-Neptune orbiting the K dwarf TOI-4311. *MNRAS*, **549**, *stag952*.
- J184 Quinn, S. N., Rodriguez, J. E., Collins, K. A., et al. (including **Ziegler, C.**) (2026). TOI-2494 c and TOI-5143 c: A Hot Saturn and a Hot Jupiter with Interior Planetary Companions. *AJ*, **171**, 359.
- J183 Šubjak, J., Brahm, R., Lipták, J., et al. (including **Ziegler, C.**) (2026). Unraveling the brown dwarf desert: Four new discoveries and a unifying period-coded picture. *A&A*, **709**, A130.
- J182 Wilson, T. G., Simpson, A. M., Collier Cameron, A., et al. (including **Ziegler, C.**) (2026). Gas-depleted planet formation occurred in the four-planet system around the red dwarf LHS 1903. *Science*, **392**, *ead12348*.
- J181 Kendall, A., Ulmer-Moll, S., Gill, S., et al. (including **Ziegler, C.**) (2026). A 43 d transiting Neptune and two 25 d Saturns from TESS, NGTS, and ASTEP. *MNRAS*, **547**, *staf2189*.
- J180 Mihaylov, D. P., Eberhardt, J., Trifonov, T., et al. (including **Ziegler, C.**) (2026). A Warm Massive Pair of Planets around TOI-1232 Revealed with Transit-timing Variations and Doppler Spectroscopy. *AJ*, **171**, 173.
- J179 Scott, M. G., Dransfield, G., Timmermans, M., et al. (including **Ziegler, C.**) (2026). Two temperate Earth- and Neptune-sized planets orbiting fully convective M dwarfs. *MNRAS*, **547**, *stag070*.
- J178 Reinartz, Y., Jones, M. I., Brahm, R., et al. (including **Ziegler, C.**) (2026). A transiting hot Jupiter with two outer siblings orbiting an intermediate-mass post-main-sequence star. *A&A*, **706**, A20.
- J177 Espinoza-Retamal, J. I., Brahm, R., Petrovich, C., et al. (including **Ziegler, C.**) (2026). A Cold and Superpuffy Planet on a Prograde Orbit. *ApJL*, **996**, L13.
- J176 Plunkett, T., Thygesen, E., Cole, A. A., et al. (including **Ziegler, C.**) (2026). The TASSIE program II: Three close-in companions orbiting sun-like stars. *PASA*, **43**, e112.

## 2025

- J175 Eschen, Y. N. E., Wilson, T. G., Bonfanti, A., et al. (including **Ziegler, C.**) (2025). An ultra-short period super-Earth and sub-Neptune spanning the Radius Valley orbiting the kinematic thick disc star TOI-2345. *MNRAS*, **544**, 2614-2636.
- J174 Ulmer-Moll, S., Gill, S., Brahm, R., et al. (including **Ziegler, C.**) (2025). Detection and characterisation of a 106-day transiting Jupiter: TOI-2449 b/NGTS-36 b. *A&A*, **703**, A258.
- J173 Murgas, F., Pallé, E., Suárez Mascareño, A., et al. (including **Ziegler, C.**) (2025). TOI-283 b: A transiting mini-Neptune in a 17.6-day orbit discovered with TESS and ESPRESSO. *A&A*, **703**, A201.
- J172 Yalçinkaya, S., Barkaoui, K., Baştürk, Ö., et al. (including **Ziegler, C.**) (2025). TOI-1743 b, TOI-5799 b, TOI-5799 c, and TOI-6223 b: TESS discovery and validation of four super-Earth to Neptune-sized planets around M dwarfs. *A&A*, **702**, A209.
- J171 Hobson, M. J., Suárez Mascareño, A., Lovis, C., et al. (including **Ziegler, C.**) (2025). TOI-2322: Two transiting rocky planets close to the stellar rotation period and its first harmonic. *A&A*, **702**, A32.
- J170 Schulte, J., Rodriguez, J. E., Latham, D. W., et al. (including **Ziegler, C.**) (2025). Migration and Evolution of giant ExoPlanets (MEEP). II. Super-Jupiters and Lithium-rich host stars. *MNRAS*, **543**, 292-313.
- J169 Yee, S. W., Winn, J. N., Hartman, J. D., et al. (including **Ziegler, C.**) (2025). The TESS Grand Unified Hot Jupiter Survey. III. Thirty More Giant Planets. *ApJS*, **280**, 30.
- J168 Manni, F., Naponiello, L., Mancini, L., et al. (including **Ziegler, C.**) (2025). The Hot Neptune Initiative (HONEI). II. TOI-5795 b: A hot super-Neptune orbiting a metal-poor star. *A&A*, **701**, A230.
- J167 Naponiello, L., Vissapragada, S., Bonomo, A. S., et al. (including **Ziegler, C.**) (2025). The Hot-Neptune Initiative (HONEI). I. Two hot sub-Neptunes on a close-in eccentric orbit (TOI-5800 b) and a farther-out circular orbit (TOI-5817 b). *A&A*, **701**, A79.
- J166 Jenkins, S. A., Vanderburg, A., Sethi, R., et al. (including **Ziegler, C.**) (2025). An Eccentric Sub-Neptune Moving Into the Evaporation Desert. *ApJL*, **989**, L20.
- J165 Crisp, A. L., Klüter, J., Newman, M. L., et al. (including **Ziegler, C.**) (2025). The Multiband Imaging Survey for High-alpha PlanetS (MISHAPS). I. Preliminary Constraints on the Occurrence Rate of Hot Jupiters in 47 Tucanae. *AJ*, **170**, 104.
- J164 Hesse, K., Mireles, I., Bouchy, F., et al. (including **Ziegler, C.**) (2025). HD 35843: A Sun-like Star Hosting a Long-period Sub-Neptune and Inner Super-Earth. *AJ*, **170**, 108.
- J163 Frensch, Y. G. C., Bouchy, F., Lo Curto, G., et al. (including **Ziegler, C.**) (2025). Three hot Jupiters transiting K-dwarfs with significant heavy element masses. *A&A*, **700**, A118.
- J162 Lockley, I. S., Armstrong, D. J., Fernández Fernández, J., et al. (including **Ziegler, C.**) (2025). The TOI-1117 multiplanetary system: 3 sub-Neptunes, 1 in both the Neptunian Desert and Radius Valley. *MNRAS*, **541**, 919-938.
- J161 Vowell, N., Rodriguez, J. E., Latham, D. W., et al. (including **Ziegler, C.**) (2025). Eleven New Transiting Brown Dwarfs and Very-low-mass Stars from TESS. *AJ*, **170**, 68.
- J160 Janó Muñoz, C., Hooton, M. J., Pedersen, P. P., et al. (including **Ziegler, C.**) (2025). TOI-2407 b: a warm Neptune in the desert. *MNRAS*, **541**, 630-641.
- J159 Essack, Z., Dragomir, D., Dalba, P. A., et al. (including **Ziegler, C.**) (2025). Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. VI. Confirmation of a Long-period Giant Planet Discovered with a Single TESS Transit. *AJ*, **170**, 41.
- J158 Alqasim, A., Hirano, T., Hori, Y., et al. (including **Ziegler, C.**) (2025). Investigating the eccentricity distribution of transiting, long-period giant planets. *MNRAS*, **539**, 307-329.

- J157 Bieryla, A., Dong, J., Zhou, G., et al. (including **Ziegler, C.**) (2025). TOI-2005b: An Eccentric Warm Jupiter in Spin-orbit Alignment. *AJ*, **169**, 273.
- J156 Chaturvedi, P., Goffo, E., Gandolfi, D., et al. (including **Ziegler, C.**) (2025). TOI-3493 b: A planet with a Neptune-like density transiting a bright G0-type star. *A&A*, **697**, A169.
- J155 Stalport, M., Mortier, A., Cretignier, M., et al. (including **Ziegler, C.**) (2025). TESS and HARPS-N unveil two planets transiting TOI-1453: A super-Earth and one of the lowest mass sub-Neptunes. *A&A*, **696**, A86.
- J154 Distler, A., Soares-Furtado, M., Vanderburg, A., et al. (including **Ziegler, C.**) (2025). TESS Hunt for Young and Maturing Exoplanets (THYME). XII. A Young Mini-Neptune on the Upper Edge of the Radius Valley in the Hyades Cluster. *AJ*, **169**, 166.
- J153 Barkaoui, K., Korth, J., Gaidos, E., et al. (including **Ziegler, C.**) (2025). TOI-2015 b: A sub-Neptune in strong gravitational interaction with an outer non-transiting planet. *A&A*, **695**, A281.
- J152 Rodrigues, J., Barros, S. C. C., Santos, N. C., et al. (including **Ziegler, C.**) (2025). TOI-512: Super-Earth transiting a K-type star discovered by TESS and ESPRESSO. *A&A*, **695**, A237.
- J151 Matson, R. A., Gore, R., Howell, S. B., et al. (including **Ziegler, C.**) (2025). Demographics of M Dwarf Binary Exoplanet Hosts Discovered by TESS. *AJ*, **169**, 76.
- J150 Rodríguez Martínez, R., Eastman, J. D., Collins, K. A., et al. (including **Ziegler, C.**) (2025). Discovery and Characterization of an Eccentric, Warm Saturn Transiting the Solar Analog TOI-4994. *AJ*, **169**, 72.
- J149 Thomas, L., Hébrard, G., Kellermann, H., et al. (including **Ziegler, C.**) (2025). TOI-5108 b and TOI 5786 b: Two transiting sub-Saturns detected and characterized with TESS, MaHPS, and SOPHIE. *A&A*, **694**, A143.
- J148 Heidari, N., Hébrard, G., Martioli, E., et al. (including **Ziegler, C.**) (2025). Characterization of seven transiting systems, including four warm Jupiters from SOPHIE and TESS. *A&A*, **694**, A36.
- J147 Vítková, M., Brahm, R., Trifonov, T., et al. (including **Ziegler, C.**) (2025). TOI-4504: Exceptionally Large Transit Timing Variations Induced by Two Resonant Warm Gas Giants in a Three-planet System. *ApJL*, **978**, L22.
- J146 Šubjak, J., Gandolfi, D., Goffo, E., et al. (including **Ziegler, C.**) (2025). TOI-2458 b: A mini-Neptune consistent with in situ hot Jupiter formation. *A&A*, **693**, A235.

## 2024

- J145 Ehrhardt, J., Thomas, L., Kellermann, H., et al. (including **Ziegler, C.**) (2024). Confirmation of four hot Jupiters detected by TESS using follow-up spectroscopy from MaHPS at Wendelstein together with NEID and TRES. *A&A*, **692**, A220.
- J144 Nies, M., Mireles, I., Bouchy, F., et al. (including **Ziegler, C.**) (2024). HD 21520 b: a warm sub-Neptune transiting a bright G dwarf. *MNRAS*, **534**, 3744-3760.
- J143 Castro-González, A., Lillo-Box, J., Armstrong, D. J., et al. (including **Ziegler, C.**) (2024). TOI-5005 b: A super-Neptune in the savanna near the ridge. *A&A*, **691**, A233.
- J142 Peláez-Torres, A., Esparza-Borges, E., Pallé, E., et al. (including **Ziegler, C.**) (2024). Validation of up to seven TESS planet candidates through multi-colour transit photometry using MuSCAT2 data. *A&A*, **690**, A62.
- J141 Carleo, I., Barrágan, O., Persson, C. M., et al. (including **Ziegler, C.**) (2024). Mass determination of two Jupiter-sized planets orbiting slightly evolved stars: TOI-2420 b and TOI-2485 b. *A&A*, **690**, A18.
- J140 Henderson, B. A., Casewell, S. L., Jordán, A., et al. (including **Ziegler, C.**) (2024). TOI-2490b - the most eccentric brown dwarf transiting in the brown dwarf desert. *MNRAS*, **533**, 2823-2842.
- J139 Nabbie, E., Huang, C. X., Burt, J. A., et al. (including **Ziegler, C.**) (2024). Surviving in the Hot-Neptune Desert: The Discovery of the Ultrahot Neptune TOI-3261b. *AJ*, **168**, 132.

- J138 Gupta, A. F., Millholland, S. C., Im, H., et al. (including **Ziegler, C.**) (2024). A hot-Jupiter progenitor on a super-eccentric retrograde orbit. *Nature*, **632**, 50-54.
- J137 Hacker, A., Díaz, R. F., Armstrong, D. J., et al. (including **Ziegler, C.**) (2024). TOI-2374 b and TOI-3071 b: two metal-rich sub-Saturns well within the Neptunian desert. *MNRAS*, **532**, 1612-1634.
- J136 Šubjak, J., Latham, D. W., Quinn, S. N., et al. (including **Ziegler, C.**) (2024). Evolution of BD-14 3065b (TOI-4987b) from giant planet to brown dwarf as possible evidence of deuterium burning at old stellar ages. *A&A*, **688**, A120.
- J135 Thao, P. C., Mann, A. W., Barber, M. G., et al. (including **Ziegler, C.**) (2024). TESS Hunt for Young and Maturing Exoplanets (THYME). X. A Two-planet System in the 210 Myr MELANGE-5 Association. *AJ*, **168**, 41.
- J134 Schulte, J., Rodriguez, J. E., Bieryla, A., et al. (including **Ziegler, C.**) (2024). Migration and Evolution of giant Exoplanets (MEEP). I. Nine Newly Confirmed Hot Jupiters from the TESS Mission. *AJ*, **168**, 32.
- J133 Montalto, M., Greco, N., Biazzo, K., et al. (including **Ziegler, C.**) (2024). The GAPS programme at TNG. LVII. TOI-5076b: A warm sub-Neptune planet orbiting a thin-to-thick-disk transition star in a wide binary system. *A&A*, **687**, A226.
- J132 Timmermans, M., Dransfield, G., Gillon, M., et al. (including **Ziegler, C.**) (2024). TOI-4336 A b: A temperate sub-Neptune ripe for atmospheric characterization in a nearby triple M-dwarf system. *A&A*, **687**, A48.
- J131 Hori, Y., Fukui, A., Hirano, T., et al. (including **Ziegler, C.**) (2024). The Discovery and Follow-up of Four Transiting Short-period Sub-Neptunes Orbiting M Dwarfs. *AJ*, **167**, 289.
- J130 Tey, E., Shporer, A., Lin, Z., et al. (including **Ziegler, C.**) (2024). GJ 238 b: A 0.57 Earth Radius Planet Orbiting an M2.5 Dwarf Star at 15.2 pc. *AJ*, **167**, 283.
- J129 Battley, M. P., Collins, K. A., Ulmer-Moll, S., et al. (including **Ziegler, C.**) (2024). NGTS-30b/TOI-4862b: An 1 Gyr old 98-day transiting warm Jupiter. *A&A*, **686**, A230.
- J128 Hord, B. J., Kempton, E. M. R., Evans-Soma, T. M., et al. (including **Ziegler, C.**) (2024). Identification of the Top TESS Objects of Interest for Atmospheric Characterization of Transiting Exoplanets with JWST. *AJ*, **167**, 233.
- J127 Mallorquín, M., Lodieu, N., Béjar, V. J. S., et al. (including **Ziegler, C.**) (2024). TOI-1135 b: A young hot Saturn-size planet orbiting a solar-type star. *A&A*, **685**, A90.
- J126 Psaridi, A., Osborn, H., Bouchy, F., et al. (including **Ziegler, C.**) (2024). Discovery of two warm mini-Neptunes with contrasting densities orbiting the young K3V star TOI-815. *A&A*, **685**, A5.
- J125 Mistry, P., Prasad, A., Maity, M., et al. (including **Ziegler, C.**) (2024). VaTEST III: Validation of eight potential super-earths from TESS data. *PASA*, **41**, e030.
- J124 Page, E., Pepper, J., Wright, D., et al. (including **Ziegler, C.**) (2024). TOI-1994b: A Low-mass Eccentric Brown Dwarf Transiting A Subgiant Star. *AJ*, **167**, 109.
- J123 Jones, M. I., Reinartz, Y., Brahm, R., et al. (including **Ziegler, C.**) (2024). A long-period transiting substellar companion in the super-Jupiters to brown dwarfs mass regime and a prototypical warm-Jupiter detected by TESS. *A&A*, **683**, A192.
- J122 Capistrant, B. K., Soares-Furtado, M., Vanderburg, A., et al. (including **Ziegler, C.**) (2024). TESS Hunt for Young and Maturing Exoplanets (THYME). XI. An Earth-sized Planet Orbiting a Nearby, Solar-like Host in the 400 Myr Ursa Major Moving Group. *AJ*, **167**, 54.
- J121 Carleo, I., Malavolta, L., Desidera, S., et al. (including **Ziegler, C.**) (2024). The GAPS programme at TNG. L. TOI-4515 b: An eccentric warm Jupiter orbiting a 1.2 Gyr-old G-star. *A&A*, **682**, A135.

## 2023

- J120 Harris, M., Dragomir, D., Mireles, I., et al. (including **Ziegler, C.**) (2023). Separated Twins or Just Siblings? A Multi-planet System around an M Dwarf Including a Cool Sub-Neptune. *ApJL*, **959**, L1.
- J119 Eberhardt, J., Hobson, M. J., Henning, T., et al. (including **Ziegler, C.**) (2023). Three Warm Jupiters around Solar-analog Stars Detected with TESS. *AJ*, **166**, 271.
- J118 Mann, C. R., Dalba, P. A., Lafrenière, D., et al. (including **Ziegler, C.**) (2023). Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. III. Recovery and Confirmation of a Temperate, Mildly Eccentric, Single-transit Jupiter Orbiting TOI-2010. *AJ*, **166**, 239.
- J117 Martioli, E., Hébrard, G., de Almeida, L., et al. (including **Ziegler, C.**) (2023). TOI-1736 and TOI-2141: Two systems including sub-Neptunes around solar analogs revealed by TESS and SOPHIE. *A&A*, **680**, A84.
- J116 Hagelberg, J., Nielsen, L. D., Attia, M., et al. (including **Ziegler, C.**) (2023). TOI-858 B b: A hot Jupiter on a polar orbit in a loose binary. *A&A*, **679**, A70.
- J115 Damasso, M., Rodrigues, J., Castro-González, A., et al. (including **Ziegler, C.**) (2023). A compact multi-planet system transiting HIP 29442 (TOI-469) discovered by TESS and ESPRESSO. Radial velocities lead to the detection of transits with low signal-to-noise ratio. *A&A*, **679**, A33.
- J114 Naponiello, L., Mancini, L., Sozzetti, A., et al. (including **Ziegler, C.**) (2023). A super-massive Neptune-sized planet. *Nature*, **622**, 255-260.
- J113 Hawthorn, F., Bayliss, D., Armstrong, D. J., et al. (including **Ziegler, C.**) (2023). TOI-908: a planet at the edge of the Neptune desert transiting a G-type star. *MNRAS*, **524**, 3877-3893.
- J112 Sha, L., Vanderburg, A. M., Huang, C. X., et al. (including **Ziegler, C.**) (2023). TESS spots a mini-neptune interior to a hot saturn in the TOI-2000 system. *MNRAS*, **524**, 1113-1138.
- J111 Deeg, H. J., Georgieva, I. Y., Nowak, G., et al. (including **Ziegler, C.**) (2023). TOI-1416: A system with a super-Earth planet with a 1.07 d period. *A&A*, **677**, A12.
- J110 Lin, Z., Gan, T., Wang, S. X., et al. (including **Ziegler, C.**) (2023). Three low-mass companions around aged stars discovered by TESS. *MNRAS*, **523**, 6162-6185.
- J109 Tuson, A., Queloz, D., Osborn, H. P., et al. (including **Ziegler, C.**) (2023). TESS and CHEOPS discover two warm sub-Neptunes transiting the bright K-dwarf HD 15906. *MNRAS*, **523**, 3090-3118.
- J108 Osborn, H. P., Nowak, G., Hébrard, G., et al. (including **Ziegler, C.**) (2023). Two warm Neptunes transiting HIP 9618 revealed by TESS and Cheops. *MNRAS*, **523**, 3069-3089.
- J107 Frame, G., Armstrong, D. J., Cegla, H. M., et al. (including **Ziegler, C.**) (2023). TOI-2498 b: a hot bloated super-Neptune within the Neptune desert. *MNRAS*, **523**, 1163-1174.
- J106 Hua, X., Wang, S. X., Teske, J. K., et al. (including **Ziegler, C.**) (2023). A Transiting Super-Earth in the Radius Valley and an Outer Planet Candidate Around HD 307842. *AJ*, **166**, 32.
- J105 Mistry, P., Pathak, K., Prasad, A., et al. (including **Ziegler, C.**) (2023). VaTEST. II. Statistical Validation of 11 TESS-detected Exoplanets Orbiting K-type Stars. *AJ*, **166**, 9.
- J104 Psaridi, A., Bouchy, F., Lendl, M., et al. (including **Ziegler, C.**) (2023). Three Saturn-mass planets transiting F-type stars revealed with TESS and HARPS: TOI-615b, TOI-622b, and TOI-2641b. *A&A*, **675**, A39.
- J103 Vowell, N., Rodriguez, J. E., Quinn, S. N., et al. (including **Ziegler, C.**) (2023). HIP 33609 b: An Eccentric Brown Dwarf Transiting a  $V = 7.3$  Rapidly Rotating B Star. *AJ*, **165**, 268.
- J102 Brahm, R., Ulmer-Moll, S., Hobson, M. J., et al. (including **Ziegler, C.**) (2023). Three Long-period Transiting Giant Planets from TESS. *AJ*, **165**, 227.

- J101 Rodriguez, J. E., Quinn, S. N., Vanderburg, A., et al. (including **Ziegler, C.**) (2023). Another shipment of six short-period giant planets from TESS. *MNRAS*, **521**, 2765-2785.
- J100 Mann, C., Lafrenière, D., Dragomir, D., et al. (including **Ziegler, C.**) (2023). Validation of TOI-1221 b: A Warm Sub-Neptune Exhibiting Transit Timing Variations around a Sun-like Star. *AJ*, **165**, 217.
- J99 Kálmán, S., Derekas, A., Csizmadia, S., et al. (including **Ziegler, C.**) (2023). Discovery of a substellar companion in the TESS light curve of the  $\delta$  Scuti/ $\gamma$  Doradus hybrid pulsator HD 31221. *A&A*, **673**, L14.
- J98 Barros, S. C. C., Demangeon, O. D. S., Armstrong, D. J., et al. (including **Ziegler, C.**) (2023). The young mini-Neptune HD 207496b that is either a naked core or on the verge of becoming one. *A&A*, **673**, A4.
- J97 Yee, S. W., Winn, J. N., Hartman, J. D., et al. (including **Ziegler, C.**) (2023). The TESS Grand Unified Hot Jupiter Survey. II. Twenty New Giant Planets. *ApJS*, **265**, 1.
- J96 Oddo, D., Dragomir, D., Brandeker, A., et al. (including **Ziegler, C.**) (2023). Characterization of a Set of Small Planets with TESS and CHEOPS and an Analysis of Photometric Performance. *AJ*, **165**, 134.
- J95 Heitzmann, A., Zhou, G., Quinn, S. N., et al. (including **Ziegler, C.**) (2023). TOI-4562b: A Highly Eccentric Temperate Jupiter Analog Orbiting a Young Field Star. *AJ*, **165**, 121.
- J94 Tey, E., Huang, C. X., Kunimoto, M., et al. (including **Ziegler, C.**) (2023). TESS Discovery of Twin Planets near 2:1 Resonance around Early M Dwarf TOI 4342. *AJ*, **165**, 93.
- J93 Wood, M. L., Mann, A. W., Barber, M. G., et al. (including **Ziegler, C.**) (2023). TESS Hunt for Young and Maturing Exoplanets (THYME). IX. A 27 Myr Extended Population of Lower Centaurus Crux with a Transiting Two-planet System. *AJ*, **165**, 85.
- J92 Magliano, C., Covone, G., Dobal, R., et al. (including **Ziegler, C.**) (2023). A systematic validation of hot Neptunes in TESS data. *MNRAS*, **519**, 1562-1577.
- J91 Tofflemire, B. M., Kraus, A. L., Mann, A. W., et al. (including **Ziegler, C.**) (2023). A Low-mass, Pre-main-sequence Eclipsing Binary in the 40 Myr Columba Association-Fundamental Stellar Parameters and Modeling the Effect of Star Spots. *AJ*, **165**, 46.
- J90 del Ser, D., Fors, O., del Alcázar, M., et al. (including **Ziegler, C.**) (2023). TFAW survey II: six newly validated planets and 13 planet candidates from K2. *MNRAS*, **518**, 669-690.
- J89 El Mufti, M., Plavchan, P. P., Isaacson, H., et al. (including **Ziegler, C.**) (2023). TOI 560: Two Transiting Planets Orbiting a K Dwarf Validated with iSHELL, PFS, and HIRES RVs. *AJ*, **165**, 10.

## 2022

- J88 Murgas, F., Nowak, G., Masseron, T., et al. (including **Ziegler, C.**) (2022). HD 20329b: An ultra-short-period planet around a solar-type star found by TESS. *A&A*, **668**, A158.
- J87 Caciapuoti, L., Inno, L., Covone, G., et al. (including **Ziegler, C.**) (2022). TESS discovery of a super-Earth and two sub-Neptunes orbiting the bright, nearby, Sun-like star HD 22946. *A&A*, **668**, A85.
- J86 Persson, C. M., Georgieva, I. Y., Gandolfi, D., et al. (including **Ziegler, C.**) (2022). TOI-2196 b: Rare planet in the hot Neptune desert transiting a G-type star. *A&A*, **666**, A184.
- J85 Dransfield, G., Triaud, A. H. M. J., Guillot, T., et al. (including **Ziegler, C.**) (2022). HD 28109 hosts a trio of transiting Neptunian planets including a near-resonant pair, confirmed by ASTEP from Antarctica. *MNRAS*, **515**, 1328-1345.
- J84 Almenara, J. M., Bonfils, X., Otegi, J. F., et al. (including **Ziegler, C.**) (2022). GJ 3090 b: one of the most favourable mini-Neptune for atmospheric characterisation. *A&A*, **665**, A91.
- J83 Barragán, O., Armstrong, D. J., Gandolfi, D., et al. (including **Ziegler, C.**) (2022). The young HD 73583 (TOI-560) planetary system: two 10- $M_{\oplus}$  mini-Neptunes transiting a 500-Myr-old, bright, and active K dwarf. *MNRAS*, **514**, 1606-1627.

- J82 Giacalone, S., Dressing, C. D., García Muñoz, A., et al. (including **Ziegler, C.**) (2022). HD 56414 b: A Warm Neptune Transiting an A-type Star. *ApJL*, **935**, L10.
- J81 Vach, S., Quinn, S. N., Vanderburg, A., et al. (including **Ziegler, C.**) (2022). TOI-712: A System of Adolescent Mini-Neptunes Extending to the Habitable Zone. *AJ*, **164**, 71.
- J80 Yee, S. W., Winn, J. N., Hartman, J. D., et al. (including **Ziegler, C.**) (2022). The TESS Grand Unified Hot Jupiter Survey. I. Ten TESS Planets. *AJ*, **164**, 70.
- J79 Hord, B. J., Colón, K. D., Berger, T. A., et al. (including **Ziegler, C.**) (2022). The Discovery of a Planetary Companion Interior to Hot Jupiter WASP-132 b. *AJ*, **164**, 13.
- J78 Christiansen, J. L., Bhure, S., Zink, J. K., et al. (including **Ziegler, C.**) (2022). Scaling K2. V. Statistical Validation of 60 New Exoplanets From K2 Campaigns 2-18. *AJ*, **163**, 244.
- J77 Christian, S., Vanderburg, A., Becker, J., et al. (including **Ziegler, C.**) (2022). A Possible Alignment Between the Orbits of Planetary Systems and their Visual Binary Companions. *AJ*, **163**, 207.
- J76 Salama, M., **Ziegler, C.**, Baranec, C., et al. (2022). An Adaptive Optics Census of Companions to Northern Stars Within 25 pc with Robo-AO. *AJ*, **163**, 200.
- J75 Serrano, L. M., Gandolfi, D., Mustill, A. J., et al. (including **Ziegler, C.**) (2022). A low-eccentricity migration pathway for a 13-h-period Earth analogue in a four-planet system. *Nature Astronomy*, **6**, 736-750.
- J74 Dong, J., Huang, C. X., Zhou, G., et al. (including **Ziegler, C.**) (2022). NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star. *ApJL*, **926**, L7.
- J73 Giacalone, S., Dressing, C. D., Hedges, C., et al. (including **Ziegler, C.**) (2022). Validation of 13 Hot and Potentially Terrestrial TESS Planets. *AJ*, **163**, 99.

## 2021

- J72 Martin, D. V., El-Badry, K., Hodžić, V. K., et al. (including **Ziegler, C.**) (2021). TOI-1259Ab - a gas giant planet with 2.7 per cent deep transits and a bound white dwarf companion. *MNRAS*, **507**, 4132-4148.
- J71 **Ziegler, C.**, Tokovinin, A., Latiolais, M.<sup>†</sup>, Briceño, C., Law, N., & Mann, A. W. (2021). SOAR TESS Survey. II. The Impact of Stellar Companions on Planetary Populations. *AJ*, **162**, 192.
- J70 Osborn, A., Armstrong, D. J., Cale, B., et al. (including **Ziegler, C.**) (2021). TOI-431/HIP 26013: a super-Earth and a sub-Neptune transiting a bright, early K dwarf, with a third RV planet. *MNRAS*, **507**, 2782-2803.
- J69 Gan, T., Bedell, M., Wang, S. X., et al. (including **Ziegler, C.**) (2021). HD 183579b: a warm sub-Neptune transiting a solar twin detected by TESS. *MNRAS*, **507**, 2220-2240.
- J68 Burt, J. A., Dragomir, D., Mollière, P., et al. (including **Ziegler, C.**) (2021). TOI-1231 b: A Temperate, Neptune-sized Planet Transiting the Nearby M3 Dwarf NLTT 24399. *AJ*, **162**, 87.
- J67 Moutou, C., Almenara, J. M., Hébrard, G., et al. (including **Ziegler, C.**) (2021). TOI-1296b and TOI-1298b observed with TESS and SOPHIE: two hot Saturn-mass exoplanets with different densities around metal-rich stars. *A&A*, **653**, A147.
- J66 Adams, E. R., Jackson, B., Johnson, S., et al. (including **Ziegler, C.**) (2021). Ultra-short-period Planets in K2. III. Neighbors are Common with 13 New Multiplanet Systems and 10 Newly Validated Planets in Campaigns 0-8 and 10. *PSJ*, **2**, 152.
- J65 Cointepas, M., Almenara, J. M., Bonfils, X., et al. (including **Ziegler, C.**) (2021). TOI-269 b: an eccentric sub-Neptune transiting a M2 dwarf revisited with ExTrA. *A&A*, **650**, A145.
- J64 Osborn, H. P., Armstrong, D. J., Adibekyan, V., et al. (including **Ziegler, C.**) (2021). A hot mini-Neptune in the radius valley orbiting solar analogue HD 110113. *MNRAS*, **502**, 4842-4857.

- J63 Addison, B. C., Wright, D. J., Nicholson, B. A., et al. (including **Ziegler, C.**) (2021). TOI-257b (HD 19916b): a warm sub-saturn orbiting an evolved F-type star. *MNRAS*, **502**, 3704-3722.
- J62 Rodriguez, J. E., Quinn, S. N., Zhou, G., et al. (including **Ziegler, C.**) (2021). TESS Delivers Five New Hot Giant Planets Orbiting Bright Stars from the Full-frame Images. *AJ*, **161**, 194.
- J61 Dawson, R. I., Huang, C. X., Brahm, R., et al. (including **Ziegler, C.**) (2021). Precise Transit and Radial-velocity Characterization of a Resonant Pair: The Warm Jupiter TOI-216c and Eccentric Warm Neptune TOI-216b. *AJ*, **161**, 161.
- J60 **Ziegler, C.**, Law, N., Baranec, C., Riddle, R., & Tokovinin, A. (2021). Robo-AO and SOAR High-resolution Surveys of Exoplanet Hosting Stars. *Frontiers in Astronomy and Space Sciences*, **8**, 625230.
- J59 Carmichael, T. W., Quinn, S. N., Zhou, G., et al. (including **Ziegler, C.**) (2021). TOI-811b and TOI-852b: New Transiting Brown Dwarfs with Similar Masses and Very Different Radii and Ages from the TESS Mission. *AJ*, **161**, 97.
- J58 Daylan, T., Pínglé, K., Wright, J., et al. (including **Ziegler, C.**) (2021). TESS Discovery of a Super-Earth and Three Sub-Neptunes Hosted by the Bright, Sun-like Star HD 108236. *AJ*, **161**, 85.
- J57 Sha, L., Huang, C. X., Shporer, A., et al. (including **Ziegler, C.**) (2021). TOI-954 b and K2-329 b: Short-period Saturn-mass Planets that Test whether Irradiation Leads to Inflation. *AJ*, **161**, 82.
- J56 Newton, E. R., Mann, A. W., Kraus, A. L., et al. (including **Ziegler, C.**) (2021). TESS Hunt for Young and Maturing Exoplanets (THYME). IV. Three Small Planets Orbiting a 120 Myr Old Star in the Pisces-Eridanus Stream. *AJ*, **161**, 65.
- J55 Weiss, L. M., Dai, F., Huber, D., et al. (including **Ziegler, C.**) (2021). The TESS-Keck Survey. II. An Ultra-short-period Rocky Planet and Its Siblings Transiting the Galactic Thick-disk Star TOI-561. *AJ*, **161**, 56.
- J54 Ment, K., Irwin, J., Charbonneau, D., et al. (including **Ziegler, C.**) (2021). TOI 540 b: A Planet Smaller than Earth Orbiting a Nearby Rapidly Rotating Low-mass Star. *AJ*, **161**, 23.
- J53 Waalkes, W. C., Berta-Thompson, Z. K., Collins, K. A., et al. (including **Ziegler, C.**) (2021). TOI 122b and TOI 237b: Two Small Warm Planets Orbiting Inactive M Dwarfs Found by TESS. *AJ*, **161**, 13.
- J52 Zhou, G., Quinn, S. N., Irwin, J., et al. (including **Ziegler, C.**) (2021). Two Young Planetary Systems around Field Stars with Ages between 20 and 320 Myr from TESS. *AJ*, **161**, 2.
- J51 Luque, R., Serrano, L. M., Molaverdikhani, K., et al. (including **Ziegler, C.**) (2021). A planetary system with two transiting mini-Neptunes near the radius valley transition around the bright M dwarf TOI-776. *A&A*, **645**, A41.

## 2020

- J50 Curtis, J. L., Agüeros, M. A., Matt, S. P., et al. (including **Ziegler, C.**) (2020). When Do Stalled Stars Resume Spinning Down? Advancing Gyrochronology with Ruprecht 147. *ApJ*, **904**, 140.
- J49 Bouma, L. G., Hartman, J. D., Brahm, R., et al. (including **Ziegler, C.**) (2020). Cluster Difference Imaging Photometric Survey. II. TOI 837: A Young Validated Planet in IC 2602. *AJ*, **160**, 239.
- J48 Brahm, R., Nielsen, L. D., Wittenmyer, R. A., et al. (including **Ziegler, C.**) (2020). TOI-481 b and TOI-892 b: Two Long-period Hot Jupiters from the Transiting Exoplanet Survey Satellite. *AJ*, **160**, 235.
- J47 Davis, A. B., Wang, S., Jones, M., et al. (including **Ziegler, C.**) (2020). TOI 564 b and TOI 905 b: Grazing and Fully Transiting Hot Jupiters Discovered by TESS. *AJ*, **160**, 229.
- J46 Nowak, G., Luque, R., Parviainen, H., et al. (including **Ziegler, C.**) (2020). The CARMENES search for exoplanets around M dwarfs. Two planets on opposite sides of the radius gap transiting the nearby M dwarf LTT 3780. *A&A*, **642**, A173.
- J45 Mireles, I., Shporer, A., Grieves, N., et al. (including **Ziegler, C.**) (2020). TOI 694b and TIC 220568520b: Two Low-mass Companions near the Hydrogen-burning Mass Limit Orbiting Sun-like Stars. *AJ*, **160**, 133.

- J44 Gilbert, E. A., Barclay, T., Schlieder, J. E., et al. (including **Ziegler, C.**) (2020). The First Habitable-zone Earth-sized Planet from TESS. I. Validation of the TOI-700 System. *AJ*, **160**, 116.
- J43 Rodríguez Martínez, R., Gaudi, B. S., Rodríguez, J. E., et al. (including **Ziegler, C.**) (2020). KELT-25 b and KELT-26 b: A Hot Jupiter and a Substellar Companion Transiting Young A Stars Observed by TESS. *AJ*, **160**, 111.
- J42 Teske, J., Díaz, M. R., Luque, R., et al. (including **Ziegler, C.**) (2020). TESS Reveals a Short-period Sub-Neptune Sibling (HD 86226c) to a Known Long-period Giant Planet. *AJ*, **160**, 96.
- J41 Rowden, P., Borkovits, T., Jenkins, J. M., et al. (including **Ziegler, C.**) (2020). TIC 278956474: Two Close Binaries in One Young Quadruple System Identified by TESS. *AJ*, **160**, 76.
- J40 Armstrong, D. J., Lopez, T. A., Adibekyan, V., et al. (including **Ziegler, C.**) (2020). A remnant planetary core in the hot-Neptune desert. *Nature*, **583**, 39-42.
- J39 Carmichael, T. W., Quinn, S. N., Mustill, A. J., et al. (including **Ziegler, C.**) (2020). Two Intermediate-mass Transiting Brown Dwarfs from the TESS Mission. *AJ*, **160**, 53.
- J38 Rizzuto, A. C., Newton, E. R., Mann, A. W., et al. (including **Ziegler, C.**) (2020). TESS Hunt for Young and Maturing Exoplanets (THYME). II. A 17 Myr Old Transiting Hot Jupiter in the Sco-Cen Association. *AJ*, **160**, 33.
- J37 Cloutier, R., Eastman, J. D., Rodríguez, J. E., et al. (including **Ziegler, C.**) (2020). A Pair of TESS Planets Spanning the Radius Valley around the Nearby Mid-M Dwarf LTT 3780. *AJ*, **160**, 3.
- J36 Nielsen, L. D., Brahm, R., Bouchy, F., et al. (including **Ziegler, C.**) (2020). Three short-period Jupiters from TESS. HIP 65Ab, TOI-157b, and TOI-169b. *A&A*, **639**, A76.
- J35 Kostov, V. B., Orosz, J. A., Feinstein, A. D., et al. (including **Ziegler, C.**) (2020). TOI-1338: TESS' First Transiting Circumbinary Planet. *AJ*, **159**, 253.
- J34 Eisner, N. L., Barragán, O., Aigrain, S., et al. (including **Ziegler, C.**) (2020). Planet Hunters TESS I: TOI 813, a subgiant hosting a transiting Saturn-sized planet on an 84-day orbit. *MNRAS*, **494**, 750-763.
- J33 Gan, T., Shporer, A., Livingston, J. H., et al. (including **Ziegler, C.**) (2020). LHS 1815b: The First Thick-disk Planet Detected by TESS. *AJ*, **159**, 160.
- J32 Jordán, A., Brahm, R., Espinoza, N., et al. (including **Ziegler, C.**) (2020). TOI-677b: A Warm Jupiter (P = 11.2 days) on an Eccentric Orbit Transiting a Late F-type Star. *AJ*, **159**, 145.
- J31 Lamman, C., Baranec, C., Berta-Thompson, Z. K., et al. (including **Ziegler, C.**) (2020). Robo-AO M-dwarf Multiplicity Survey: Catalog. *AJ*, **159**, 139.
- J30 Astudillo-Defru, N., Cloutier, R., Wang, S. X., et al. (including **Ziegler, C.**) (2020). A hot terrestrial planet orbiting the bright M dwarf L 168-9 unveiled by TESS. *A&A*, **636**, A58.
- J29 Díaz, M. R., Jenkins, J. S., Gandolfi, D., et al. (including **Ziegler, C.**) (2020). TOI-132 b: A short-period planet in the Neptune desert transiting a V = 11.3 G-type star. *MNRAS*, **493**, 973-985.
- J28 Lendl, M., Bouchy, F., Gill, S., et al. (including **Ziegler, C.**) (2020). TOI-222: a single-transit TESS candidate revealed to be a 34-d eclipsing binary with CORALIE, EulerCam, and NGTS. *MNRAS*, **492**, 1761-1769.
- J27 Jenkins, J. S., Díaz, M. R., Kurtovic, N. T., et al. (including **Ziegler, C.**) (2020). An ultrahot Neptune in the Neptune desert. *Nature Astronomy*, **4**, 1148-1157.
- J26 Espinoza, N., Brahm, R., Henning, T., et al. (including **Ziegler, C.**) (2020). HD 213885b: a transiting 1-d-period super-Earth with an Earth-like composition around a bright (V = 7.9) star unveiled by TESS. *MNRAS*, **491**, 2982-2999.
- J25 **Ziegler, C.**, Tokovinin, A., Briceño, C., Mang, J.<sup>‡</sup>, Law, N., & Mann, A. W. (2020). SOAR TESS Survey. I. Sculpting of TESS Planetary Systems by Stellar Companions. *AJ*, **159**, 19.

## 2019

- J24 Quinn, S. N., Becker, J. C., Rodriguez, J. E., et al. (including **Ziegler, C.**) (2019). Near-resonance in a System of Sub-Neptunes from TESS. *AJ*, **158**, 177.
- J23 Vanderburg, A., Huang, C. X., Rodriguez, J. E., et al. (including **Ziegler, C. A.**) (2019). TESS Spots a Compact System of Super-Earths around the Naked-eye Star HR 858. *ApJL*, **881**, L19.
- J22 Bowler, B. P., Hinkley, S., **Ziegler, C.**, et al. (2019). The Elusive Majority of Young Moving Groups. I. Young Binaries and Lithium-rich Stars in the Solar Neighborhood. *ApJ*, **877**, 60.
- J21 Rodriguez, J. E., Quinn, S. N., Huang, C. X., et al. (including **Ziegler, C.**) (2019). An Eccentric Massive Jupiter Orbiting a Subgiant on a 9.5-day Period Discovered in the Transiting Exoplanet Survey Satellite Full Frame Images. *AJ*, **157**, 191.
- J20 Jones, M. I., Brahm, R., Espinoza, N., et al. (including **Ziegler, C.**) (2019). HD 2685 b: a hot Jupiter orbiting an early F-type star detected by TESS. *A&A*, **625**, A16.
- J19 Chené, A. N., St-Louis, N., Moffat, A. F. J., et al. (including **Ziegler, C. A.**) (2019). Investigating the origin of the spectral line profiles of the Hot Wolf-Rayet Star WR 2. *MNRAS*, **484**, 5834-5844.

## 2018

- J18 **Ziegler, C.**, Law, N. M., Baranec, C., et al. (2018). Measuring the Recoverability of Close Binaries in Gaia DR2 with the Robo-AO Kepler Survey. *AJ*, **156**, 259.
- J17 **Ziegler, C.**, Law, N. M., Baranec, C., et al. (2018). Robo-AO Kepler Survey. V. The Effect of Physically Associated Stellar Companions on Planetary Systems. *AJ*, **156**, 83.
- J16 Howard, W. S., Tilley, M. A., Corbett, H., et al. (including **Ziegler, C.**) (2018). The First Naked-eye Superflare Detected from Proxima Centauri. *ApJL*, **860**, L30.
- J15 **Ziegler, C.**, Law, N. M., Baranec, C., et al. (2018). Robo-AO Kepler Survey. IV. The Effect of Nearby Stars on 3857 Planetary Candidate Systems. *AJ*, **155**, 161.
- J14 Howard, W. S., Law, N. M., **Ziegler, C. A.**, Baranec, C., & Riddle, R. (2018). Laser-only Adaptive Optics Achieves Significant Image Quality Gains Compared to Seeing-limited Observations over the Entire Sky. *AJ*, **155**, 59.
- J13 Hillenbrand, L. A., Zhang, C., Riddle, R. L., et al. (including **Ziegler, C.**) (2018). Robo-AO Discovery and Basic Characterization of Wide Multiple Star Systems in the Pleiades, Praesepe, and NGC 2264 Clusters. *AJ*, **155**, 51.

## 2017

- J12 Schonhut-Stasik, J. S., Baranec, C., Huber, D., et al. (including **Ziegler, C.**) (2017). Robo-AO Kepler Asteroseismic Survey. I. Adaptive Optics Imaging of 99 Asteroseismic Kepler Dwarfs and Subgiants. *ApJ*, **847**, 97.
- J11 Adams, E. R., Jackson, B., Endl, M., et al. (including **Ziegler, C.**) (2017). Ultra-short-period Planets in K2 with Companions: A Double Transiting System for EPIC 220674823. *AJ*, **153**, 82.
- J10 **Ziegler, C.**, Law, N. M., Morton, T., et al. (2017). Robo-AO Kepler Planetary Candidate Survey. III. Adaptive Optics Imaging of 1629 Kepler Exoplanet Candidate Host Stars. *AJ*, **153**, 66.
- J9 Atkinson, D., Baranec, C., **Ziegler, C.**, Law, N., Riddle, R., & Morton, T. (2017). Probability of the Physical Association of 104 Blended Companions to Kepler Objects of Interest Using Visible and Near-infrared Adaptive Optics Photometry. *AJ*, **153**, 25.

## 2016

- J8 Crossfield, I. J. M., Ciardi, D. R., Petigura, E. A., et al. (including **Ziegler, C.**) (2016). 197 Candidates and 104 Validated Planets in K2's First Five Fields. *ApJS*, **226**, 7.

- J7 Baranec, C., **Ziegler, C.**, Law, N. M., et al. (2016). Robo-AO Kepler Planetary Candidate Survey. II. Adaptive Optics Imaging of 969 Kepler Exoplanet Candidate Host Stars. *AJ*, **152**, 18.
- J6 Schlieder, J. E., Crossfield, I. J. M., Petigura, E. A., et al. (including **Ziegler, C.**) (2016). Two Small Temperate Planets Transiting Nearby M Dwarfs in K2 Campaigns 0 and 1. *ApJ*, **818**, 87.

## 2015

- J5 David, T. J., Stauffer, J., Hillenbrand, L. A., et al. (including **Ziegler, C.**) (2015). HII 2407: An Eclipsing Binary Revealed By K2 Observations of the Pleiades. *ApJ*, **814**, 62.
- J4 Fulton, B. J., Collins, K. A., Gaudi, B. S., et al. (including **Ziegler, C.**) (2015). KELT-8b: A Highly Inflated Transiting Hot Jupiter and a New Technique for Extracting High-precision Radial Velocities from Noisy Spectra. *ApJ*, **810**, 30.
- J3 **Ziegler, C.**, Law, N. M., Baranec, C., Riddle, R. L., & Fuchs, J. T. (2015). Multiplicity of the Galactic Senior Citizens: A High-resolution Search for Cool Subdwarf Companions. *ApJ*, **804**, 30.

## 2014

- J2 Law, N. M., Morton, T., Baranec, C., et al. (including **Ziegler, C.**) (2014). Robotic Laser Adaptive Optics Imaging of 715 Kepler Exoplanet Candidates Using Robo-AO. *ApJ*, **791**, 35.

## 2013

- J1 Krungleviciute, V., **Ziegler, C. A.**, Banjara, S. R., Yudasaka, M., Iijima, S., & Migone, A. D. (2013). Neon and CO<sub>2</sub> Adsorption on Open Carbon Nanohorns. *Langmuir*, **29**, 9388-9397.

## Published Corrections

- E3 Sha, L., Vanderburg, A. M., Huang, C. X., et al. (including **Ziegler, C.**) (2023). Correction to: TESS spots a mini-neptune interior to a hot saturn in the TOI-2000 system. *MNRAS*, **526**, 2440-2440.
- E2 Naponiello, L., Mancini, L., Sozzetti, A., et al. (including **Ziegler, C.**) (2023). Author Correction: A super-massive Neptune-sized planet. *Nature*, **623**, E6-E6.
- E1 Jenkins, J. S., Díaz, M. R., Kurtovic, N. T., et al. (including **Ziegler, C.**) (2020). Author Correction: An ultrahot Neptune in the Neptune desert. *Nature Astronomy*, **4**, 1202-1202.

## Accepted Journal Articles

- A1 Lillo-Box, J., Cifuentes, C., Balsalobre-Ruza, O., et al. (including **Ziegler, C.**) (2026). CAFE follow-up of TESS hot Jupiter candidates left behind: I. Five newly confirmed planets and a false positive. *A&A*, accepted for publication. [arXiv:2607.19756](https://arxiv.org/abs/2607.19756).

## Conference Proceedings and Abstracts

- C5 Baranec, C., Riddle, R., Ou, J., et al. (including **Ziegler, C.**) (2026). Robo-AO-2: entering the era of automated science operations, hybrid wavefront sensing, and adaptive secondary integration. *Proc. SPIE; Adaptive Optics Systems X*.
- C4 Howard, W. S., Tilley, M. A., Corbett, H. T., et al. (including **Ziegler, C. A.**) (2018). Evryscope Detection of the First Proxima Superflare: Impacts on the Atmosphere and Habitability of Proxima b. *Comparative Climatology of Terrestrial Planets III: From Stars to Surfaces*, 2065, 2039 (conference abstract).
- C3 Law, N. M., **Ziegler, C.**, & Tokovinin, A. (2016). SRAO: the first southern robotic AO system. *Proc. SPIE*, **9907**, 99070K.
- C2 **Ziegler, C.**, Law, N. M., Baranec, C., et al. (2016). The Robo-AO KOI survey: laser adaptive optics imaging of every Kepler exoplanet candidate. *Proc. SPIE*, **9909**, 99095U.

- C1 **Ziegler, C.**, Law, N. M., & Tokovinin, A. (2016). SRAO: optical design and the dual-knife-edge WFS. *Proc. SPIE*, **9909**, 99093Z.

## Preprints and Submitted Manuscripts

---

- P10 Schulte, J., Rodriguez, J. E., Latham, D. W., et al. (including **Ziegler, C.**) (2026). Migration and Evolution of giant ExoPlanets (MEEP). III. Twenty-Nine Giant Planets from the TESS Mission. Preprint. [arXiv:2609.38468](#).
- P9 Cadieux, C., Skinner, B. N., Bouchy, F., et al. (including **Ziegler, C.**) (2026). A temperate sub-Neptune transiting the M4 Dwarf TOI-210 identified by NIRPS and TESS – Uncovering hidden M-dwarf planetary systems in the near-infrared. Submitted to A&A; under minor revision. [arXiv:2609.35747](#).
- P8 **Ziegler, C.** (2026). Gravitas: A Browser-Based Astrophysics Laboratory for Prediction, Measurement, and Discovery. Preprint. [arXiv:2609.19327](#).
- P7 **Ziegler, C.**, Tokovinin, A., Briceño, C., Law, N. M., Mann, A. W., & Batson, S.<sup>†</sup> (2026). SOAR TESS Survey III: Mapping the Stellar Companion Deficit around TESS Planet Candidates. Submitted to AJ. [arXiv:2608.26250](#).
- P6 Weldon, G. C., Yee, S. W., Hansen, B. M. S., et al. (including **Ziegler, C.**) (2026). Discovery of an Inflated Hot Neptune and Its Formation from Jovian Mass Loss. Preprint. [arXiv:2607.01315](#).
- P5 Rojas, F. I., Brahm, R., Jordán, A., et al. (including **Ziegler, C.**) (2026). Two warm sub-Saturn mass planets identified from the TESS Full Frame Images. Preprint. [arXiv:2602.16646](#).
- P4 Frommer, A. C., Eastman, J. D., Ciardi, D. R., et al. (including **Ziegler, C.**) (2025). A Paradigm Shift in Exoplanet False Positive Identification with HIP-44302. Preprint. [arXiv:2511.21817](#).
- P3 Anderson, D. R., Vines, J. I., Hesse, K., et al. (including **Ziegler, C.**) (2025). NGTS-11 c: a transiting Neptune-mass planet interior to the warm Saturn NGTS-11 b. Preprint. [arXiv:2510.14083](#).
- P2 Gandolfi, D., Alnajjarine, A., Serrano, L., et al. (including **Ziegler, C.**) (2025). A four-planet system orbiting the old thick disk star TOI-1203. Preprint. [arXiv:2509.10136](#).
- P1 Chontos, A., Huber, D., Grunblatt, S. K., et al. (including **Ziegler, C.**) (2024). The TESS-Keck Survey XXI: 13 New Planets and Homogeneous Properties for 21 Subgiant Systems. Preprint. [arXiv:2402.07893](#).

## Published Scientific Software

---

- S2 **Ziegler, C.** (2026). Gravitas: an interactive astrophysics sandbox for teaching. Version 1.0.0. [Zenodo software archive](#).
- S1 Castro-González, A., Lillo-Box, J., Armstrong, D. J., et al. (including **Ziegler, C.**) (2025). TESS-cont: TESS contamination tool. [Astrophysics Source Code Library](#), [ascl:2506.004](#).

## Dissertation

---

- D1 **Ziegler, C. A.** (2018). Characterization of Exoplanets and Stellar Systems with New Robots. Ph.D. dissertation, University of North Carolina at Chapel Hill.